

Mo Kamel

Mechatronics Engineer • Automation & Mechanical Design

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PROFILE

Mechatronics engineer with hands-on experience in mechanical design (AutoCAD, Inventor, SolidWorks, Fusion 360), hardware and software testing, and structured technical documentation. Built and tested autonomous-system hardware from CAD through to integration, authored an ISO-compliant technical standard, and works well in cross-functional, process-driven teams. Based in Regensburg and motivated to support automation technology and continuous improvement in a manufacturing environment.

KEY COMPETENCIES

AutoCAD | Inventor | SolidWorks / Fusion 360 | 2D Layouts & 3D Modelling | Hardware & Software Testing | Automation & Mechatronic Systems
5S / Process Documentation | MATLAB / Simulink

PROFESSIONAL EXPERIENCE

Mechatronics Engineer • Part-time

Jul 2020 - Mar 2022

Invictus UUVs • Alexandria, EG

- Designed and modelled mechanical assemblies in CAD: an 8-thruster modular HDPE frame for **Brotta II**, an autonomous underwater vehicle (2nd Worldwide, MATE ROV 2021), iterating 2D layouts and 3D models through to a manufacturable build.
- Defined and ran the full hardware/software test methodology, covering integration, electronics integrity, and safety checks under realistic operating conditions.
- Authored and maintained the complete technical documentation (architecture, integration, test methodology, safety), scoring 101.5/100 - the only team in the competition to exceed 100 points.
- Diagnosed a seal failure one hour before the regional competition, implemented and verified a permanent fix, and sustained two days of fault-free operation.

Inspection Engineer • Full-time

Oct 2023 - Sep 2024

Alexandria Mineral Oils Co. (AMOC) • Alexandria, EG

- Authored an ISO 7751-compliant inspection and selection specification, adopted as a permanent company standard.
- Standardized an inspection workflow and kept procedures and data documented and up to date, reducing rework through clear, repeatable instructions across three production lines.

PROJECTS

Robotic Arm Digital Twin - Modelling & Monitoring

Mar 2026 - Present

M.Eng. Coursework Case Study • THD Cham

- Modelling a 6-DOF robotic arm in MATLAB/Simscape with controlled fault injection, fault- classification and Remaining-Useful-Life pipelines, and a real-time monitoring dashboard in MATLAB App Designer.

Driver Monitoring System (DMS) - In-Cabin ADAS

Sep 2022 - Jun 2023

Bachelor's Thesis - Grade A+, 1st Place Nationally

- Built and integrated a multi-sensor hardware system across a 3-node embedded stack (Jetson Nano + 2× Raspberry Pi); designed and validated a wearable PPG measurement band on 8 subjects.
- Deployed YOLOv8 phone detection (97.8 % mAP@0.5) and a MediaPipe drowsiness pipeline (97.47 % accuracy, 18-20 FPS) on resource-constrained embedded hardware.

TECHNICAL SKILLS

CAD & Design	AutoCAD	Inventor	SolidWorks	Fusion 360
Automation & Systems	Mechatronic Integration	HW / SW Testing	5S	Cleanroom-ready
Modelling & Sim.	MATLAB	Simulink	Simscape	
Programming	Python	C / C++	Git / GitHub Actions	
Perception & AI	OpenCV	YOLOv8	TensorFlow	
Embedded HW	Jetson Nano	Raspberry Pi	Arduino	PLCIMU
Languages	English C2	German B1→B2	French B1	Arabic (native)

EDUCATION

M.Eng. Mechatronics & Cyber-Physical Systems	Mar 2026 - Sep 2027
Technische Hochschule Deggendorf • Cham, DE	
◦ Focus: autonomous systems, digital twins, predictive maintenance, AI/ML pipelines, cyber-physical integration.	
B.Sc. Mechanical Engineering - Mechatronics	Sep 2018 - Jul 2023
Arab Academy for Science, Technology & Maritime Transport • Alexandria, EG	

AWARDS

2nd Place Worldwide - MATE ROV Competition	Aug 2021
20 international teams	
1st Place Regionally - MATE ROV Competition	Jun 2021
27 national teams	
1st Place Nationally - GPAct Talent Expo (Driver Monitoring System)	Jul 2023
Swift Act • 15 teams	

LEADERSHIP & VOLUNTEERING

Chief of Organizers & Team Guides Manager	Jul - Sep 2024
IOI - International Olympiad in Informatics • Alexandria, EG • \$3.2M event	
◦ Directed operations for 1,000+ international participants across logistics, accommodation, sponsor coordination (Huawei, JetBrains, Jane Street, TikTok, Wincent), and crisis management; led 216 organizers selected from 860+ candidates.	
Further: President, Rotaract Club of Alexandria New Era (2021-2022) • EOI Chief of Operations • EGOI Organizer (2025-2026).	